

KNN

↑

$k=3, 5, 7, 9, 11, \dots$?

↓

How to find Best value of k ?

↑

classification

$2 \rightarrow Y$
 $1 \rightarrow N$

comparing

$k=3$

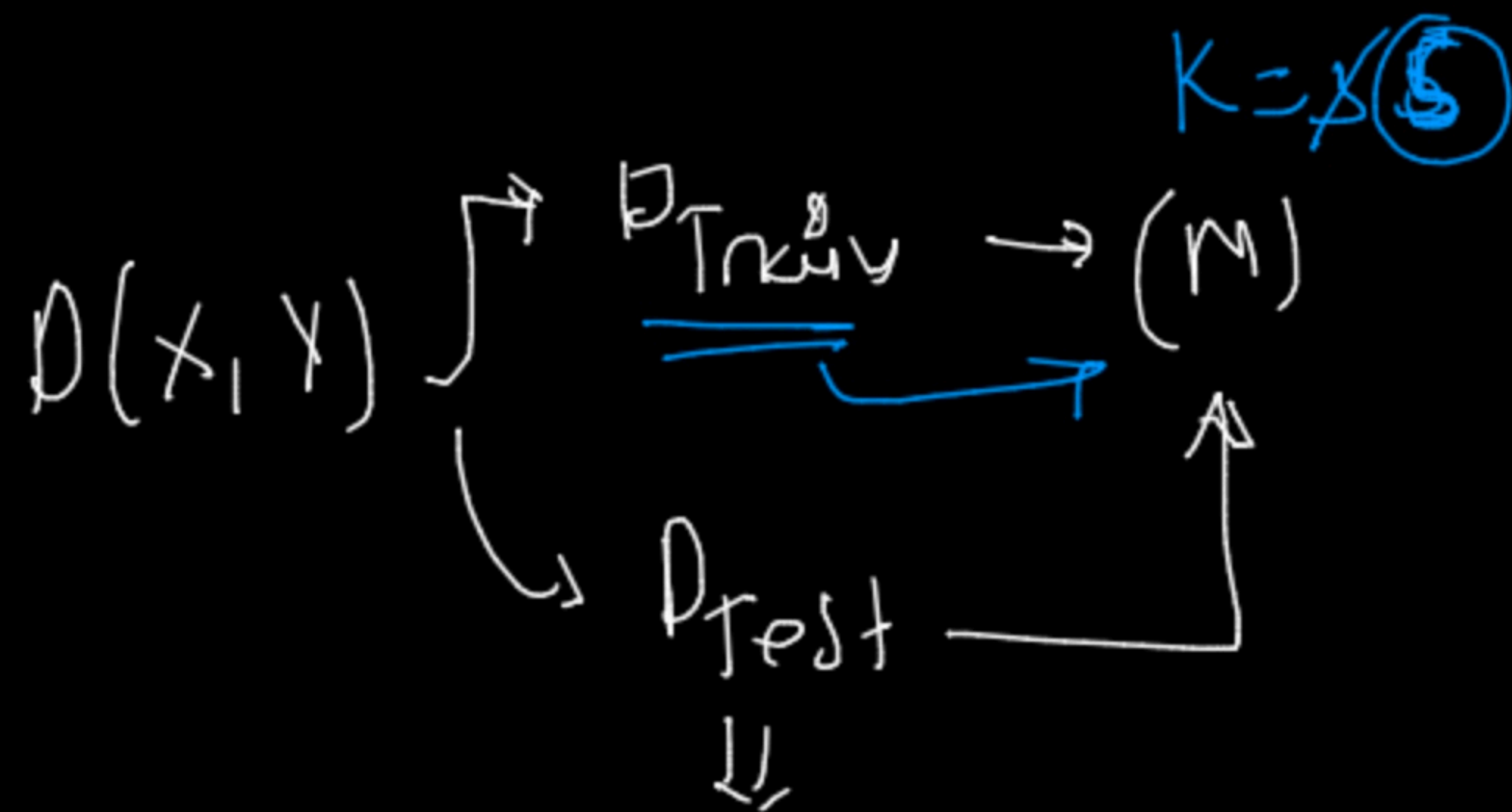
50%
50%

to find k

$1 \rightarrow Y$
 $1 \rightarrow N$

$k=3, 5, 7, 9$

odd



x	y		
x1	0	0	✓
x2	0	0	✓
x3	1	1	✓
x5	0	1	✗
x4	0	0	✗

$M()$

40%

$\frac{2}{5}$

$\rightarrow 0.4$

$k = [3, 5, 7, 9]$

accuracy = [40, 60, 70, 55]

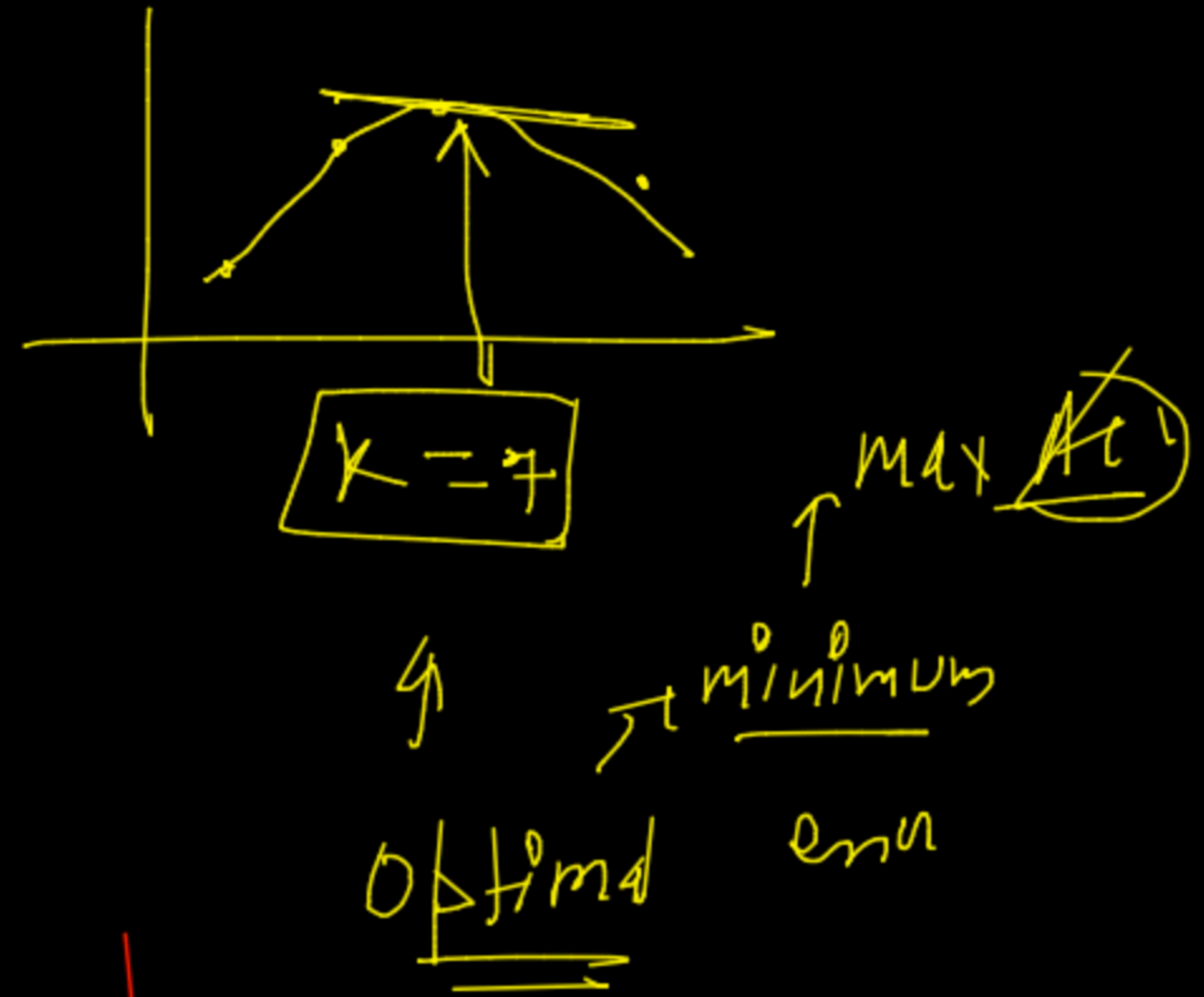
for i in ~~kns~~

¹⁵
 $knn = KNN(M=i)$

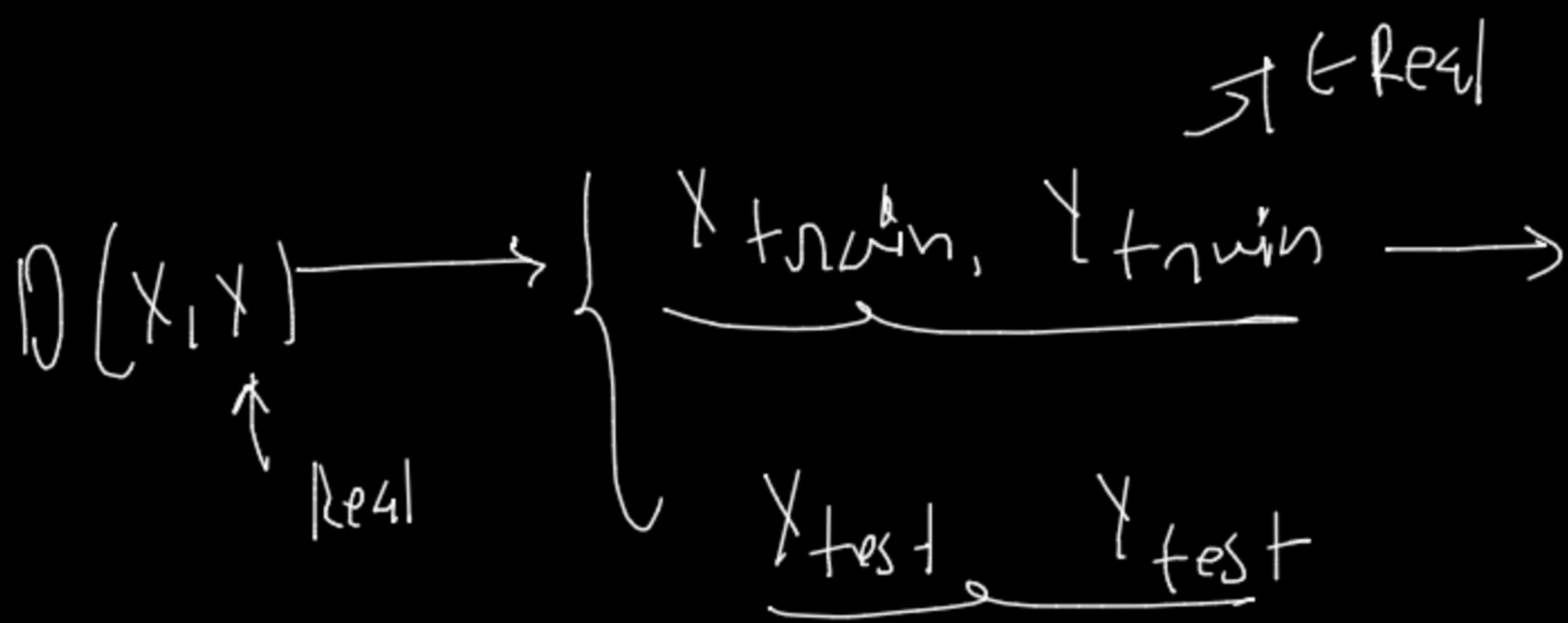
$knn.fit(D_{train})$

$y_{pred} = knn.predict(D_{test})$

acc = $accuracy(y_{pred}, y_{Actual})$



X	Y
X1	110
X2	120
X3	100
X4	
X5	



$\frac{100 + 100 + 100}{2} \Rightarrow \frac{160}{3} \approx 53.3$

X	Y	
X1	350	160
X2	340	180
X3	120	120

Real

MSE ↓ =

MSE ↑ Bad

↑ High



Classification Algorithms will work in diffn

(KNN)

$D(X, Y)$
1000
{ Yes, No }
900 100

→ imbalanced dataset

→ solution → Balance dataset

(model)

M()

will struggle to learn
the pattern →

How?

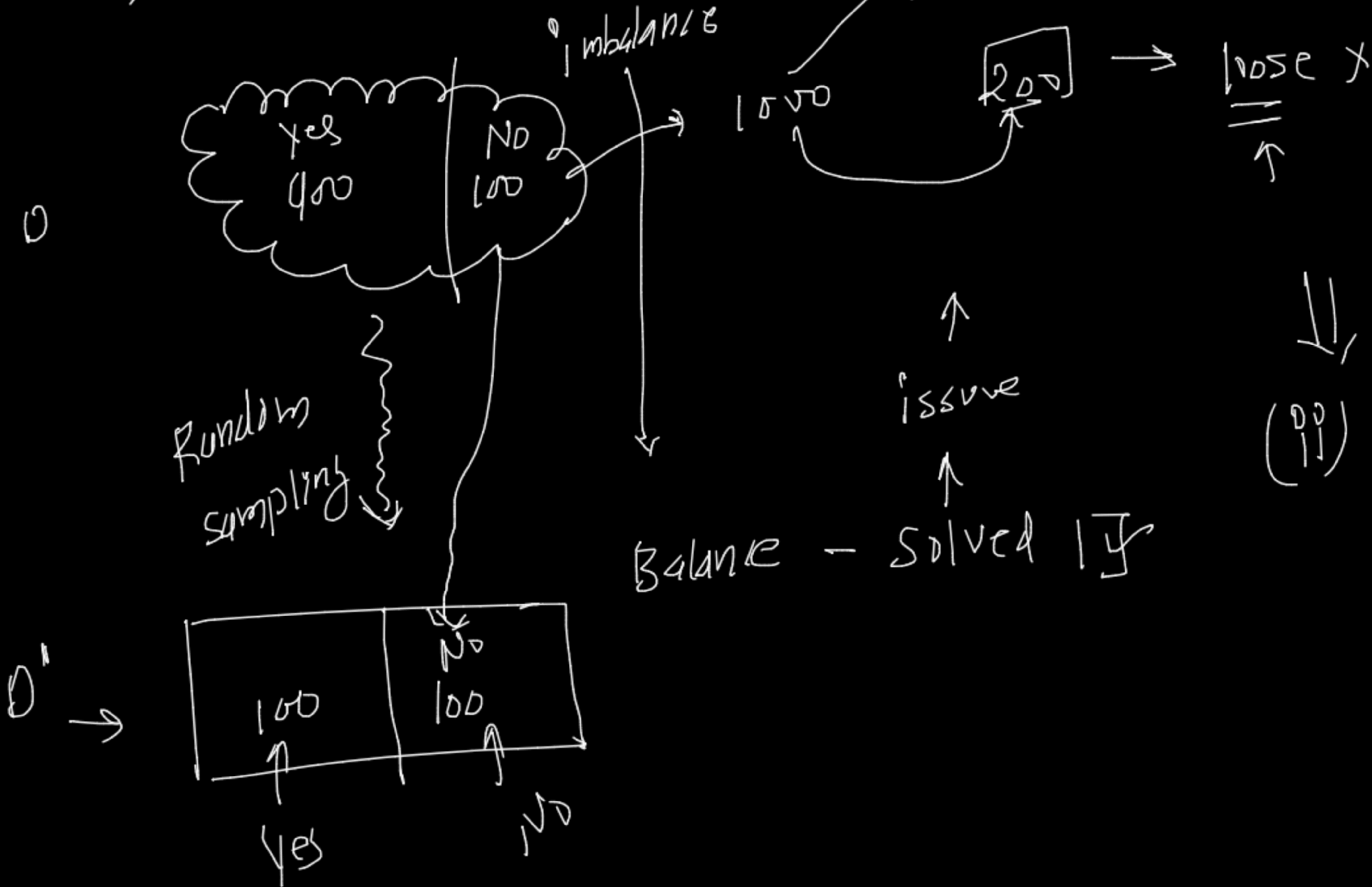
(i) oversampling

(ii) undersampling

imbalanced

Book
1000
T1 T2
400
(T1)
100
(T2)

→ Undersampling



↓
(00
11) oversampling

